

Diffusion - MNIST Summary(1FA006)

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The Jupyter notebook and the plots can be found in this [GitHub repository](#).

The aim of this assignment was to build a diffusion model to generate synthetic MNIST data using `denoising_diffusion_pytorch` package.

1 Methodology

The model setup was provided in the assignment template. The images were normalized to fall in the range $[0, 1]$. The noise prediction network is a U-Net with a base channel width of 32, scaled up across three levels (`dim_mults = (1, 2, 5)`), operating on grayscale images.

The diffusion process runs for $T = 1000$ forward timesteps, with random sampling down to 250 steps for faster inference. Training used AdamW at a learning rate of 10^{-4} , a batch size of 256, and ran for 120 epochs. I increased the batch size from 128 to 256 because the loss was less jittery by doing so. I also added a cosine `beta_schedule` in the Diffusion block to have less dramatic noising throughout the timesteps.

The part left to implement was the training loop itself. Each batch of images is passed directly to the `diffusion` object, which internally handles sampling a random timestep, adding the corresponding noise, and computing the MSE between the predicted and actual noise. A validation pass over the test set ran at the end of each epoch.

2 Results

The training converged around epoch 100 and I stopped the training at epoch 64. The sampled images look like they belong to MNIST dataset, except for a few artifact-y samples. However, the generation of particular digits was more favored by the model when running the inference. But, all digits from 0-9 made an appearance at least once in a batch of 64 generated images.

3 Challenges encountered

Since the template provided us with almost all of the code, I didn't have to exert myself much for this assignment. I tried a bunch of hyperparameter combinations so that took a lot of time. In the end, I only changed the batch size and learning rate, keeping the rest unchanged. I doubled the U-Net dimensions to 64 and I did not notice huge difference between the generated samples.